IDENTIFYING INFORMATION:

NAME: Mrenna, Stephen

PERSISTENT IDENTIFIER (PID): https://orcid.org/0000-0001-8731-160x

POSITION TITLE: Senior Scientist

PRIMARY ORGANIZATION AND LOCATION: Fermi National Accelerator Laboratory, administered by Fermi Forward Discovery Group, Batavia, IL, United States

Professional Preparation:

ORGANIZATION AND LOCATION	DEGREE (if applicable)	RECEIPT DATE	FIELD OF STUDY
University of California, Davis, Davis, CA, United States	Postdoctoral Fellow	09/1998 - 10/2001	Particle physics
Argonne National Laboratory, Lemont, IL, USA	Postdoctoral Fellow	09/1995 - 08/1998	Particle physics
California Institute of Technology, Pasadena, CA, USA	Postdoctoral Fellow	01/1993 - 09/1995	Particle physics
The Johns Hopkins University, Baltimore, MD, USA	PHD	08/1986 - 01/1993	Physics
University of Notre Dame, Notre Dame, IL, USA	BS	08/1982 - 05/1986	Physics

Appointments and Positions

2012 - present Senior Scientist, Fermi National Accelerator Laboratory, administered by Fermi Forward Discovery Group, Batavia, IL, United States

2006 - 2012 Scientist I, Fermi National Accelerator Laboratory, Batavia, IL, USA

2001 - 2006 Associate Scientist, Fermi National Accelerator Laboratory, Batavia, IL, USA

Products

1. Mrenna S, Skands P. Automated Parton-Shower Variations in Pythia 8. Phys. Rev. D. 2016; 94(7):074005. Available from: https://doi.org/10.1103/PhysRevD.94.074005 DOI: 10.1103/PhysRevD.94.074005
2. Bierlich C, Others. A comprehensive guide to the physics and usage of PYTHIA 8.3. arXiv [hep-ph]. 2022. Available from: http://dx.doi.org/10.21468/SciPostPhysCodeb.8 DOI: 10.21468/SciPostPhysCodeb.8
3. Bierlich C, Ilten P, Menzo T, Mrenna S, Szewc M, Wilkinson M, Youssef A, Zupan J. Reweighting Monte Carlo Predictions and Automated Fragmentation Variations in Pythia 8. arXiv [hep-ph]. 2023. Available from: http://arxiv.org/abs/2308.13459
4. Menzo T, Roman A, Gleyzer S, Matchev K, Fleming G, Höche S, Mrenna S, Shyamsundar P. HEPTAPOD: Orchestrating High Energy Physics Workflows Towards Autonomous Agency. arXiv [hep-ph]. 2025. Available from: http://arxiv.org/abs/2512.15867

5. Assi B, Bierlich C, Ilten P, Menzo T, Mrenna S, Szewc M, Wilkinson M, Youssef A, Zupan J. Characterizing the hadronization of parton showers using the HOMER method. SciPost Phys.. 2025; 19:125. Available from: http://dx.doi.org/10.21468/SciPostPhys.19.5.125 DOI: 10.21468/SciPostPhys.19.5.125
6. Al Kadhim A, Prosper H, Mrenna S. Bayesian optimization of pythia8 tunes. Phys. Rev. D. 2025; 112(5):056012. Available from: http://dx.doi.org/10.1103/1hn2-hz4q DOI: 10.1103/1hn2-hz4q
7. Assi B, Bierlich C, Ilten P, Menzo T, Mrenna S, Szewc M, Wilkinson M, Youssef A, Zupan J. Post-hoc reweighting of hadron production in the Lund string model. SciPost Phys.. 2025; 19(4):104. Available from: http://dx.doi.org/10.21468/SciPostPhys.19.4.104 DOI: 10.21468/SciPostPhys.19.4.104
8. Bierlich C, Ilten P, Menzo T, Mrenna S, Szewc M, Wilkinson M, Youssef A, Zupan J. Describing hadronization via histories and observables for Monte-Carlo event reweighting. SciPost Phys.. 2025; 18(2):054. Available from: http://dx.doi.org/10.21468/SciPostPhys.18.2.054 DOI: 10.21468/SciPostPhys.18.2.054
9. Bierlich C, Ilten P, Menzo T, Mrenna S, Szewc M, Wilkinson M, Youssef A, Zupan J. Reweighting Monte Carlo predictions and automated fragmentation variations in Pythia 8. SciPost Phys.. 2024; 16(5):134. Available from: http://dx.doi.org/10.21468/SciPostPhys.16.5.134 DOI: 10.21468/SciPostPhys.16.5.134
10. Li A, Macridin A, Mrenna S, Spentzouris P. Simulating scalar field theories on quantum computers with limited resources. Phys. Rev. A. 2023; 107(3):032603. Available from: http://dx.doi.org/10.1103/PhysRevA.107.032603 DOI: 10.1103/PhysRevA.107.032603

Certification:

I understand that I have been designated as a covered individual by the Federal funding agency.

I certify to the best of my knowledge and belief that the information contained in this Biographical Sketch Form is true, complete, and accurate. I understand that any false, fictitious, or fraudulent information, misrepresentations, half-truths, or omissions of any material fact, may subject me to criminal, civil or administrative penalties for fraud, false statements, false claims or otherwise. (18 U.S.C. §§ 1001 and 287, and 31 U.S.C. 3729-3733 and 3801-3812). I further understand and agree that (1) the statements and representations made herein are material to DOE's funding decision, and (2) I have a responsibility to update the disclosures during the project period of the award should circumstances change which impact the responses provided above.

I also certify that, at the time of submission, I am not a party in a malign foreign talent recruitment program. I further understand should I take action to involve myself with a Malign Foreign Talent Recruitment Program during the period of performance of the award, I must notify the recipient's Authorized Agent immediately, but no later than five business days of taking such action and immediately recuse myself from all DOE awards.

Certified by Mrenna, Stephen in SciENcv on 2026-04-14 10:54:26